

Validação de IA para o Projeto BusFlow

Guia de validação de Random Forest e SageMaker com links para execução dos testes.

Random Forest (Scikit-Learn)

Passo 1 - Criar dataset simples. Pandas:

https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html

Passo 2 - Preparar ambiente. Google Colab: <https://colab.research.google.com/>

Passo 3 - Treinar modelo.

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

Passo 4 - Definir parâmetros. <https://scikit-learn.org/stable/modules/ensemble.html#forest>

Passo 5 - Avaliar métricas. https://scikit-learn.org/stable/modules/model_evaluation.html

Passo 6 - Salvar modelo. <https://joblib.readthedocs.io/>

Passo 7 - Enviar para S3.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/upload-objects.html>

Passo 8 - Criar Lambda. <https://docs.aws.amazon.com/lambda/>

Passo 9 - Criar regra de alerta.

Passo 10 - Enviar SNS. <https://docs.aws.amazon.com/sns/>

Amazon SageMaker

Passo 1 - Validar acesso. <https://console.aws.amazon.com/sagemaker/>

Passo 2 - Criar dataset.

Passo 3 - Upload S3.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/upload-objects.html>

Passo 4 - Training Job. <https://docs.aws.amazon.com/sagemaker/latest/dg/train-model.html>

Passo 5 - Definir variável alvo.

Passo 6 - Autopilot.

<https://docs.aws.amazon.com/sagemaker/latest/dg/autopilot-automate-model-development.html>

Passo 7 - Avaliar métricas.

Passo 8 - Criar endpoint.

<https://docs.aws.amazon.com/sagemaker/latest/dg/realtime-endpoints.html>

Passo 9 - Integrar Lambda.

<https://docs.aws.amazon.com/sagemaker/latest/dg/realtime-endpoints-test-endpoints.html>

Passo 10 - SNS. <https://docs.aws.amazon.com/sns/>